Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/24/2012
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Project Title:	Proteomic aided investigation of the mechanistic basis for HZE-induced cognitive impairment and the development of diagnostic biomarkers		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
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COI Name (Institution):	Drake, Richard (Eastern Virginia Medical School) Lonart, Gyorgy (Eastern Virginia Medical School) Sanford, Larry (Eastern Virginia Medical School)		
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	One of the main concerns that NASA has about the long-term health of the astronauts who go to Mars is the impact that cosmic radiation will have on brain function. Cosmic radiation consists of charged particles, known as Hze particles. The biological effects of Hze particles are dependent upon how much energy the particles deposit when they interact with tissue, which is referred to as LET. At present we only have superficial knowledge of how some high LET radiations inhibit brain function. Our current data suggest that brain function is inhibited due to radiation-induced death of brain cells and the inhibition of the brain's signaling pathways. However, the relative importance of each of these processes at different LETs is unclear. Such information is not only vital to the development of risk assessment models, but may provide an opportunity to develop medical treatments that can slow down or even stop the progression of radiation-induced brain impairment. The central objectives of our proposed studies address Research questions 29e, 29h and 29i outlined in the current Bioastronautic Roadmap. We specifically propose to: 1) identify the high-LET radiations that have the greatest impact on the ability of animals to recall imprinted memories and to learn new memory skills; 2) use PET imaging to identify the regions of the brain that are functionally inactivated when brain function is inhibited by Hze radiation; 3) establish the relative importance of radiation-induced cell death and signal malfunction in the inhibition of brain function after exposure to radiations with low, medium and high-LETs; 4) develop a blood-based assay that can be used to monitor the astronauts for early signs of radiation-induced memory function deficit.
Task Description:	Specific Aim 1 will determine the dose required to inhibit four different memory functions in rats exposed to radiations with different LETs. The heads of rats will be exposed to graded doses of various LET radiations and the rats monitored over the following 6 months for signs of an impairment of their ability to retrieve memory and to learn new memory dependent tasks. Specific Aim 3 will determine the changes in the brain that lead to the loss of memory function in rats exposed to high-LET radiations. We shall determine the impact that radiation exposure has on the proliferation of brain cells by microscopic examination of brain slices. We shall also determine if radiation affects the ability of brain cells to release signaling proteins in response to known stimuli. Finally we shall use the new technology of proteomics, which can simultaneously measure the abundance of thousands of proteins, to identify all processes that are altered by radiation and are associated with the loss of brain function. Specific Aim 4 will use proteomics to establish the serum protein profiles of rats (who receive whole body radiation exposures) that go on to develop radiation-induced impairment of brain function. These proteins can be used as biomarkers with which to identify astronauts who have been exposed to radiation doses/energies that put them at risk of developing brain impairment. Our studies will establish the safe dose of cosmic radiation that astronauts can be exposed to without suffering from brain malfunction, which will be invaluable in deciding if the Mars mission should proceed, and if so, what additional precautions will be needed to protect the astronauts. Furthermore, our studies will provide considerable insight into how high LET radiation exposure leads to brain function impairment. These data will help develop treatments that can reduce the severity of radiation-induced brain impairment, e.g., special diets to reduce radiation-induced brain cell death, or drugs that can alter the brain's signaling pathways. Finally, our studies will identify biomarkers that can be used to identify and monitor astronauts who are at risk of developing radiation-induced brain impairment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of serum biomarkers for cognitive impairments has potential benefits in the diagnosis of neurological disorders.
	Current models predict that the astronauts on deep space missions will be exposed to ~25 cGy of Galactic cosmic radiation (GCR). The long-term consequence of exposure to such doses is largely unknown. During this grant we worked towards fulfilling our primary aims: 1. Identify the threshold dose of X-rays or HZE radiation that leads to significant deficits in neurocognitive function. 2. Identify HZE-induced changes in neuropeptide release and the CNS proteome that are associated with neurocognitive impairment. 3. Identify serum proteomics changes that can be used to monitor the emergence of HZE-induce cognitive impairment in post-mission surveillance of the astronauts. MATERIALS & METHODS. These studies were conducted using juvenile (5 week old) male Wistar rats exposed to either 1 GeV/nucleon 56Fe or 1.1 GeV/nucleon 48Ti particles at Brookhaven National Laboratory (BNL). While such young animals are not representative of the biological age of the Astronauts, they were chosen because they have a relatively high level of neurogenesis and are therefore likely to be more sensitive to HZE radiation and thus constitute the “worst case scenario.” At 3 months post exposure the ability of rats to conduct attentional set shifting and spatial memory was established. After completion of the behavioral tasks the animals were euthanized, serum and brains were recovered for subsequent proteomic analysis on the LTQ platform, or more neurotransmission release studies. RESULTS. Spatial memory impairment. Our data indicate that there is a significant level of Spatial memory impairment following exposure to low (20 cGy) doses 1 GeV/nucleon 56Fe (Britten et al., 2012). Preliminary data suggests that 10 cGy is sufficient to induced spatial memory impairment. Preliminary data also suggest that exposure to 20 cGy of 1.1 GeV/nucleon 48Ti ions also resulted in an impairment of spatial memory performance, but in contrast to 56Fe, no impairment was seen following exposure to 10 cGy 1.1 GeV/nucleon 48Ti. Thus there may be some LET dependency on

	the Threshold dose for spatial memory impairment. Importantly, there is a cohort of rats that appear to be more resistant to developing HZE-induced spatial memory impairments. Attentional Set shifting impairment. We have recently published (Lonart et al., in press) that young rats that had received 20 cGy 1 GeV/μ 56Fe cranial irradiation were incapable of completing all stages of the Attentional Set Shifting test, with many rats failing at the Reversal 1 stage. Preliminary data from rats exposed to 20 cGy of 1.1 GeV/nucleon 48Ti, suggest that they may be a higher level of impairment of this executive function than was observed with a similar dose of 1 GeV/nucleon 56Fe. Importantly, there is a cohort of rats that appear to be more resistant to developing HZE-induced attentional set shifting deficits.
Task Progress:	Changes in the neurotransmission. Exposure to 60 cGy of 1 GeV/nucleon 56Fe resulted in a significant reduction of the glutamatergic readily releasable vesicular pool (RRP) in hippocampal synaptosomes (Machida et al., 2010). This HZE-induced reduction in the glutamatergic RRP persisted for at least six months post exposure, and appears to be region specific given that the glutamatergic RRP in the associative cortex is unaffected by these HZE doses. At 90 days post exposure, 60 cGy 1 GeV/nucleon 56Fe also induced a significant decrease in the levels of NR1, NR2A, and NR2B, subunits of the glutamatergic NMDA receptor, which is critical to synaptic plasticity. In contrast, 60 cGy 1 GeV/nucleon 56Fe exposure had no impact upon the expression of the glutamatergic AMPA receptor. The observed functional impairment of glutamatergic transmission, at both the pre- (RRP) and post- (NMDA receptor) synaptic level is likely to have a significant effect on learning and memory, and offers an explanation for the loss of neurocognitive function after exposure to HZE doses that do not lead to any significant changes in neurogenesis. Hippocampal proteomic changes. We have identified (using LTQ proteomic analysis) proteomic signatures that are correlated with either a good or bad spatial memory performance status after exposure to mission relevant doses of 1 GeV/nucleon 56Fe. The emerging picture is that rats that are able to maintain good spatial memory performance are capable of “tolerating” the persistent oxidative damage, and while the normal functionality of the neurocognitive process is impacted following radiation (proteins are lost), these rats are able to, at least in part, compensate for the loss of such proteins by up-regulating other proteins that target the downstream proteins of the lost proteins. In contrast, the irradiated rats that go on to developed impaired spatial memory performance have multiple proteomic changes that do little to circumvent the lost proteins but rather lead to a highly disorganized pattern of cross-talking networks. Identification of serum biomarkers of HZE-induced cognitive impairment. We have identified changes in the serum proteome that are correlated with the absorbed dose, but more importantly identified proteins that are differentially expressed in animals that are proficient in spatial memory (Barnes’ Maze) after exposure to high (200 cGy) doses of 1 GeV/nucleon 56Fe (Britten et al., 2010). Our preliminary data suggest that there are proteins expressed in the serum that could be used as prognostic biomarkers for the development of HZE-induced spatial memory impairment SUMMARY. These studies suggest that mission-relevant doses (20 cGy) of HZE particles with LETs of 100-150 KeV/u lead to significant neurocognitive defects (spatial memory performance and attentional set shifting) in juvenile male Wistar rats. However, there is a sub-group of individuals that appears to be less susceptible to developing neurocognitive impairment. We have started to identify some of the underlying causes for this differential susceptibility to developing HZE-induced cognitive impairment, and it appears that changes in the functionality of the hippocampal synapses and also changes in the neuroproteome rather than impairment of neurogenesis may be the underlying cause of the neurocognitive impairment after low dose HZE exposures. Future studies need to determine whether more age appropriate rats show a similar susceptibility to developing HZE-induced neurocognitive impairments.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Davis LK, Johnson AM, Keeney S, Siegel A, Sanford LD, Singletary SJ, Lonart G. "Low (20 cGy) doses of 1 GeV/u (56)Fe-particle radiation lead to a persistent reduction in the spatial learning ability of rats." Radiation Research. 2012 Feb;177(2):146-51. Epub 2011 Nov 11. PubMed PMID: 22077338 , Feb-2012
Articles in Peer-reviewed Journals	Lonart G, Parris B, Johnson AM, Miles S, Sanford LD, Singletary SJ, Britten RA. "Executive function in rats is impaired by low (20 cGy) doses of 1 GeV/u (56)Fe particles." Radiation Research. 2012 Oct;178(4):289-94. Epub 2012 Aug 10. PubMed PMID: 22880624 , Oct-2012
Articles in Peer-reviewed Journals	Britten RA, Wellman LL, Sanford LD. "Progressive increase in the complexity and translatability of rodent testing to assess space-radiation induced cognitive impairment." Neurosci Biobehav Rev. 2021 Jul;126:159-74. Available online 22 March 2021. Review. https://doi.org/10.1016/j.neubiorev.2021.01.027 ; PMID: 33766676 , Jul-2021